Universidad de Oviedo
Escuela Ingenieria Informática

TRABAJO DE FIN DE GRADO GRADO EN INGENIERÍA INFORMÁTICA DEL SOFTWARE

Evaluación de la influencia de colores en interfaces en modo oscuro mediante eye-tracking

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Chapter 1

Acknowledgments

hola hola hola

Chapter 2

Abstract

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Chapter 3

Introduction

3.1 Context

Computer scientists have always been seen as "hackers" that work with Command Line Interfaces (CLI). This belief comes from a long-gone era where Graphical User Interfaces (GUI) wouldn't exist. In reality, Command Line Interfaces and Graphical User Interfaces live together in peace in the different sectors of informatics. However, lately the Graphical User Interfaces are experiencing, perhaps what can be called a 'golden era'. The evolution from the first Graphical User Interface computer, the Xerox Alto in 1973[13],to Web GUIs applications still share the same core. However, they have quickly evolved in the past years thanks to eased access to the Internet and computers. Nowadays, big companies dictate the cutting edge of technology regarding GUIs. A trend that has become obvious is the rise of Dark mode, dividing the whole user community of many applications.

3.2 Motivation

Year after year we have seen how the whole world has changed at a really high pace. Two of the fastest changing areas are Graphical User Interfaces and User Experience (UX). A long time has passed since the times when developers would consider the front end a secondary part of development.

Nowadays a new problem has become viral, not only inside the informatics world but throughout the whole spectrum of internet users: is dark mode better (from now on negative polarity) or bright mode (from now on positive polarity)? This has led many companies to embrace dark mode as its corporative colors, for example the social network X (former Twitter) shifted to negative polarity as its default mode.

There have been many studies regarding usability between both options. However, no one has actually studied the effects of the positive and negative display polarity in longer texts. That has risen the following questions:

- Does the user get more fatigued while reading in negative polarity?
- Will they read faster or slower?

- Will the user struggle to find more information?
- E-commerce websites are mainly programmed in positive polarity. Is the negative polarity really a disadvantage for the user experience while reading and analyzing a product?

All these questions will be analyzed and given the best answer possible in this thesis. A conclusion regarding long texts and display polarities will be studied and explained here.

3.3 Purpose of the thesis

The purpose of the thesis is to investigate the visual fatigue and readability of both display polarities in terms of reading long texts in web-based products. This will be achieved by using a tobii eye-tracker. This tool can give real time data of the eye, giving us objective and solid information.

Visual fatigue can be defined as a moment in which the eyes have a huge workload, and the user can perceive a sensation of discomfort. Visual fatigue can happen after a long period of time in front of a screen, but it also happens every time someone reads a text on a smaller scale. It is usually studied with questionnaires leading to subjectiveness in the analysis. However, in recent times, tools such as eye-trackers have proven a new way to study it. Symptoms of visual fatigue can be calculated as the time the user takes to read a whole paragraph and how many times the person takes to stop at each word.

3.4 Document Structure

The structure of the document has been briefly inspired by Jose Manuel Redondo's spreadsheet of final thesis for development thesis [5]. However, in order to adapt it to an investigation thesis the spreadsheet has been adapted and now doesn't follow non specific standard. This new "version" is mostly inspired by the scientific method by expanding the common topics on a scientific paper, while at the same time being mixed with the most common sections of a usual informatics thesis.

Chapter 4

State of the Art

The state of the art is the information relevant to the study that will be done. In this way, we can close the scope of what this new investigation will prove and find studies that can help us close that new research gap. As a small reminder, display polarity is the mode that will be shown on the screen, positive polarity being dark characters in bright background and negative polarity the other way around.

4.1 Previous Studies

The previous studies section, as explained above, consists of doing research in order to slowly close the research gap. This process was done by following a “baking” process, where an initial research was done on the subject in several Databases such as IEEExplore[2] or Scopus[9], refining the keywords every time to find related articles. The best keywords appeared to be:

- Display polarity
- Visual fatigue
- Reading

After reading and analyzing the first bake, a second bake of refinement and vertical expansion (analyzing cites and references of such scientific papers) was done. After erasing the unrelated articles and the ones out of scope, the following key articles were found:

4.1.1 The impact of web page text-background colour combinations on readability, retention, aesthetics and behavioural intention[4]

Table 4.1: Summary of the article by Hall and Hanna

Authors	Hall, Richard H Hanna, Patrick
Year	2004
Source	Behaviour and Information Technology
DOI	10.1080/01449290410001669932

Reason for choice

The motivation to use this paper as a reference is because of the Likert scale questionnaire that is used, giving questions that can perfectly fit with the desired study. In addition, they hand out more questions that could be applied to further explain other non-studied parts of my thesis.

Abstract

The purpose of this experiment was to examine the effect of web page text/background colour combination on readability, retention, aesthetics, and behavioural intention. One hundred and thirty-six participants studied two Web pages, one with educational content and one with commercial content, in one of four colour-combination conditions. Major findings were: (a) Colours with greater contrast ratio generally lead to greater readability; (b) colour combination did not significantly affect retention; (c) preferred colours (i.e., blues and chromatic colours) led to higher ratings of aesthetic quality and intention to purchase; and (d) ratings of aesthetic quality were significantly related to intention to purchase.

4.1.2 Study on the effects of display color mode and luminance contrast on visual fatigue. [15]

Table 4.2: Summary of the article by Xie *et al.*

Authors	Xie, Xiaojiao Song, Fanghao Liu, Yan Wang, Shurui Yu, Dong
Year	2021
Source	IEE Access
DOI	`10.1109/ACCESS.2021.3061770`

Reason for choice

This study is one of the closest to what the purpose of this thesis is related to. However, there are some differences. For example, the aim is different, and they are not focused on usability. Nevertheless, this paper has provided good information to reach the desired experiment for investigating the usability and visual fatigue. In the following illustration we can see the luminance contrast: (a) 0.969, (b) 0.935, (c) 0.855, (d) 0.868, (e) 0.725, (f) 0.469

Figure 4.1: Experimental Illustrations.

Abstract

Using electronic devices at night can easily cause visual fatigue. We investigated the conjoint effects of color mode and luminance contrast on visual fatigue and subjective preference when using electronic devices under low screen luminance and low ambient illumination at night. A multidimensional approach based on eye and subjective measures was used to test 2 color modes (dark mode, light mode) and 6 luminance contrast ratios (0.969, 0.935, 0.868, 0.855, 0.725, 0.469) in a 2×6 experimental design. We used eye movement tracking technology to collect blink rate and pupil diameters, and used the Likert scale to measure subjective visual fatigue scale and preference.

Results showed that reading in the dark mode reduced visual fatigue, as reflected by an increase in blink rate and pupil accommodation. Lower subjective visual fatigue scale and higher preference were found in the light mode due to subjects' using habits of dark texts on a light background. There was a significant negative correlation between (text-background) luminance contrast and visual fatigue, and subjects preferred higher luminance contrast. We observed the lowest visual fatigue under the luminance contrast of 0.969 in the dark mode, and the lowest subjective preference when the luminance contrast was lower than 0.725. We suggest the users should choose the dark mode to reduce visual fatigue when using electronic devices at night. These findings also provide a reference for the design of interactive interfaces such as tablets and mobile phones, and have practical implications for reducing visual fatigue.

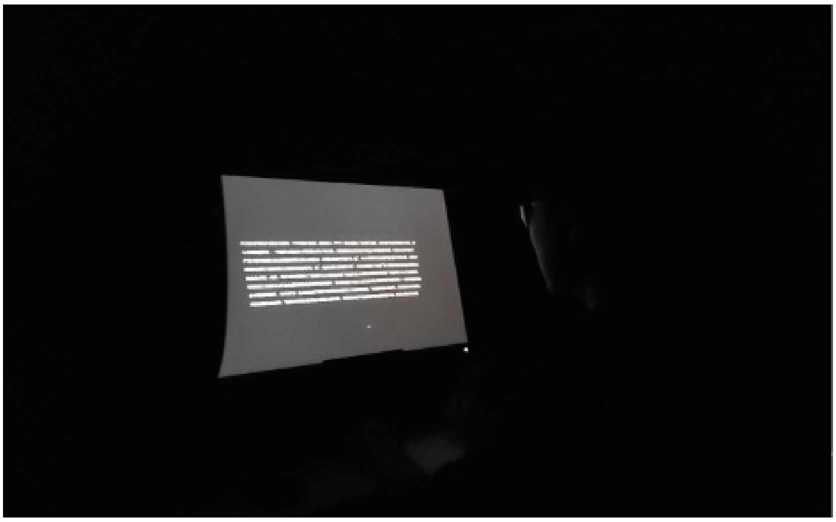

Figure 4.2: Photo of participant during the experiment

4.1.3 Web Design Attributes in Building User Trust, Satisfaction, and Loyalty for a High Uncertainty Avoidance Culture.[3]

Table 4.3: Summary of the article by Faisal *et al.*

Authors	Faisal, C M Gonzalez-Rodríguez, Martin Fernandez-Lanvinv, Daniel De Andres-Suarez, Javier
Year	2016
Source	IEEE Access
DOI	`10.1109/THMS.2016.2620901`

Reason for choice

In order to extract usability subjective data, it is necessary to use a Likert questionnaire. The display polarity questions are already selected. However, there are a few general preference questions that need to be asked. It is better to obtain well studied questions rather than using self-created ones.

Abstract

In this study, we attempt to evaluate the user preferences for web design attributes (i.e., typography, color, content quality, interactivity, and navigation) to determine the trust, satisfaction, and loyalty for uncertainty avoidance cultures. Content quality and navigation have been observed as strong factors in building user trust with e-commerce websites. In contrast, interactivity, color, and typography have been observed as strong determinants of user satisfaction. The most relevant and interesting finding is related to typography, which has been rarely discussed in e-commerce literature. A questionnaire was designed to collect data to corroborate the proposed model and hypotheses. Further more, the partial least-squares method was adopted to analyze the collected data from the students who participated in the test (n = 558). Finally, the results of this study provide strong support to the proposed model and hypotheses. Therefore, all the web design attributes were observed as important design features to develop user trust and satisfaction for uncertainty avoidance cultures. Although both factors seem to be relevant, the relationship between trust and loyalty was observed to be stronger than between satisfaction and

loyalty; thus, trust seems to be a stronger determinant of loyalty for risk/high uncertainty avoidance cultures.

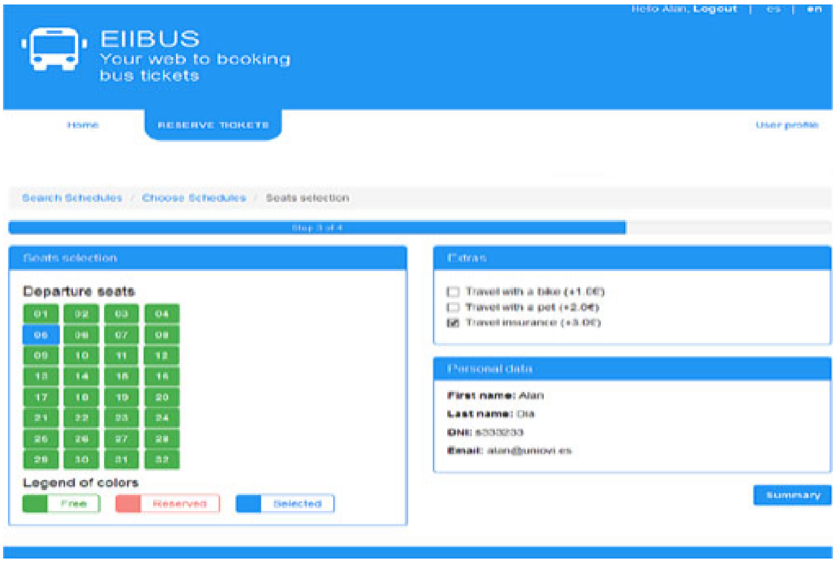

Figure 4.3: E-ticket bus website prototype used in the experiment.

4.1.4 The Effect of Display Polarity on Reading Speed and Reading Error Among Young Adults.[7]

Table 4.4: Summary of the article by Muhamad and Mokhtar

Authors	Muhamad, Nurulain Mokhtar, Nurliyana
Year	2024
Source	Journal of Health Science and Medical Research
DOI	`10.31584/jhsmr.20241095`

Reason for choice

The reason for choosing this article is the fact that, even though the study is quite similar, they are not using an eye tracker but instead measuring times by hand, hence giving a general measurement with a lot of imprecisions. However, the methodology (without considering the tools for measuring) is an acceptable one, and some of it is considered while processing the data.

Abstract

Objective: The widespread adoption of digital devices has surged, particularly since the coronavirus disease 2019 (COVID-19) pandemic. Nearly everyone now owns devices like laptops, tablets, or smartphones, offering options for light mode (positive polarity) and dark mode (negative polarity) to suit individual preferences. This study examines how display polarity affects reading performance among young adults. Material and Methods: Thirty participants engaged in a 15-minute reading task on a laptop with randomly assigned display polarities, followed by a 15-minute break before repeating the task. Results: Reading speed, measured in words per minute (wpm), differed significantly between polarities, with negative polarity yielding higher speeds (136.27 ± 25.58 wpm) compared to positive polarity (128.42 ± 19.98 wpm), Z=-2.355, p-value<0.05. However, no significant polarity-related differences were found in reading errors, including mispronunciation (p-value=0.193) or omission (p-value=0.113). Conclusion: Negative polarity displays enhanced reading performance by increasing reading speed; while reading errors remained unaffected.

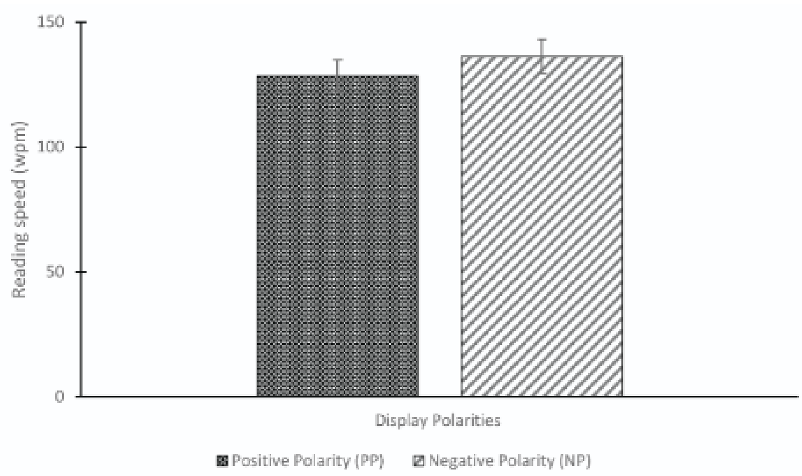

Figure 4.4: The comparison of reading speed between positive and negative display polarities.

4.1.5 Eye Tracking in Human-Computer Interaction and Usability Research: Current Status and Future Prospects [8]

Table 4.5: Summary of the article by Poole and Ball

Authors	Poole, Alex Ball, Linden J.
Year	2006
Source	Encyclopedia of Human Computer Interaction
DOI	`10.4018/978-1-59140-562-7.ch034`

Reason for choice

This article is a key reference for eye-tracking as it displays all the important information regarding the technology such as the significance of the different eye-tracking metrics like fixations, saccades, etc. It also explains how eye-trackers work. This information is the core of this study as the statistical analysis will be based on these metrics explained here.

Abstract

Eye-movement tracking is a method that is increasingly being employed to study usability issues in HCI contexts. The objectives of the present chapter are threefold. First, we introduce the reader to the basics of eye-movement technology, and also present key aspects of practical guidance to those who might be interested in using eye tracking in HCI research, whether in usability-evaluation studies, or for capturing people's eye movements as an input mechanism to drive system interaction. Second, we examine various ways in which eye movements can be systematically measured to examine interface usability. We illustrate the advantages of a range of different eye-movement metrics with reference to state-of-the-art usability research. Third, we discuss the various opportunities for eye-movement studies in future HCI research, and detail some of the challenges that need to be overcome to enable effective application of the technique in studying the complexities of advanced interactive-system use.

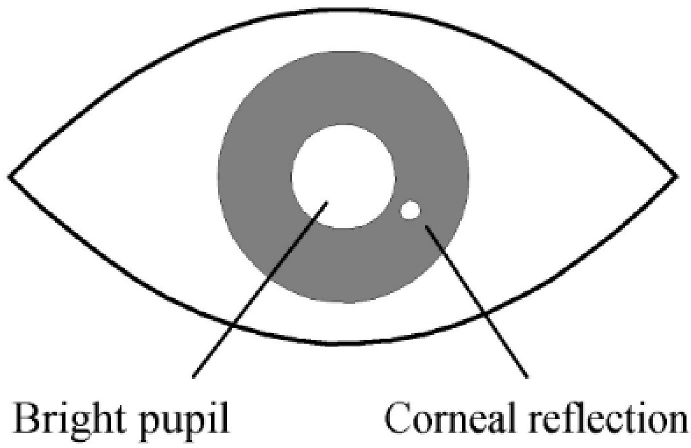

Figure 4.5: Corneal reflection and bright pupil as seen in the infrared camera such as Tobii Pro Nano [12].

4.2 Research Gap

There aren't many investigations regarding display polarity. However, the few existing ones are inconsistent. Manual measuring methods, such as taking time with a chrono, are usual procedures. Many times, the total populations could reach less than 20 people in total. Therefore, the research gap remains broad. In the case of this thesis, the investigation will be conducted for web-based wikis and product descriptions, as this can help to study the long text. It is also important to note that the aim is to give consistent results that are backed up by statistical methods and precise tools such as the Tobii eye-tracker.

Chapter 5

Hypothesis

5.1 General Objective

5.2 Specific Objectives

5.3 Hypothesis

The hypotheses to be proven in this thesis are:

H1) Users using negative polarity are less fatigued than the positive polarity users

H2) Users using negative polarity read quicker than positive polarity

H3) Users prefer negative polarity over positive polarity

Chapter 6

Methodology

6.1 Experiment Design

The experiment proposal is tightly attached to the idea of using an eye-tracking system, not only because it is an innovation that gives us more precision within the measurements but it also uniqueness surrounding the state of the art as really few experiments have used this device. To be more exact we will be using the eye tracking model Tobii Pro Nano[12]. The experiment is briefly described in this section, but it will be further explained in the chapter 7.

The user will be prompted with a text message showing the instructions. This text comes along with a previous tobii calibration that will be further explained in the section 7.2.1 and 7.2.2 respectively.

After the instructions and calibration are completed, the second phase, the actual experiment can start. There are two possible paths, negative display polarity path or positive display polarity path. Both affect the instruction texts as long as the screenshots shown to the user. The wiki text was shown always the first being followed by the product description. The user has as much time as they want. However, it wasn't possible to return to previous pages.

Once the user finished reading the third phase would start, a questionnaire regarding preference will be shown, it also contains a small set of questions about the texts to ensure the user has been reading carefully.

In addition, the webpages chosen for the experiment are `wikipedia.org` and `amazon.es`. These two webpages will be further explained in the 7.2.3 section.

6.2 Population of Study

In order to have a good experiment, not only is it important to have a consistent and well-prepared process but also have a focus population of study. The decision was to check for university students as this was the easiest and most accessible population given the situation. Therefore, the objective age was young people between 18 to 30 years old.

It is also worth mentioning that the objective was to find the usual user on the internet. For that an average user isn't a computer scientist as they can be specialized on web design or other similar fields. This point made us go to the faculty of Economics as it was a larger faculty and the students can be considered an "average" user.

The objective was to have 80 persons in total 20 women and 20 men divided evenly between the negative and positive display polarity. However, due to the lack of collaboration of the people and the tendency to decline collaboration in the experiment, this was later reduced to 34 in total.

To summarize, it can be concluded that the conditions and population of the study were:

Table 6.1: Distribution of participants by gender, polarity, and age.

Number of partic.	*Neg. Pol.*		*Pos. Pol.*		***TOTAL***		***Age***
	Obj.	Act.	Obj.	Act.	Obj.	Act.	
Men	20	7	20	7	40	14	[18,30]
Women	20	10	20	10	40	20	
Total	40	17	40	17	80	34	

6.3 Tools

In order to accomplish the success of the experiment there is a need of some tools. They will be further explained why they are important ant what purpose do they serve. There is one hardware tool requirement and 3 software tools that are used to collect data and process it.

6.3.1 Tobii Pro Nano

The tobii pro nano it's an eye-tracker currently discontinued by the owners (Tobii)[12] that is still to its day a great tool to collect user data. It's worth mentioning that this tool is quite expensive and to use it the University of Oviedo has lent me one unit. Always with the surveillance of my tutors. The tobii pro nano is the lightest and most portable unit offered, however it has some limitations that won't be a problem in our case as the limitations are for precise operations.

Figure 6.1: Tobii Pro Nano Device[12]

Eye Tracker Technology Breakdown

The eye tracker works in an amazing way, the process is a combination of high-quality cameras. Using what's commonly called Pupil Center Corneal Reflection (PCCR). In the case of tobii pro nano it uses a patented variant of PCCR (US Patent US7,572,008).[10]

The PCCR is a mix between "illuminators" that is components that are almost infrared sensors that create such light on the eyes. Then different sets of cameras take pictures of the eye patterns and reflections. That is sent to algorithms that detect those patterns and reflections creating a detailed 3D model of the eye position and its gazes [10].

6.3.2 Tobii Pro Lab

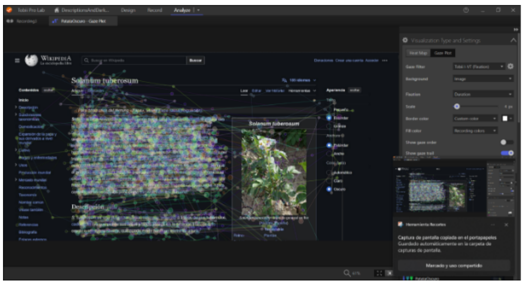

Figure 6.2: Tobii Pro Lab Software[11]

The eye-tracking device connects to a software application called "Tobii Pro Lab". This application aims to ease the use, collection and analysis of the gaze data. Curiously this is a negative display polarity program. It is needed to have a license in order to use it. However, a 30 day trial is provided which has been enough toextract the data, change or create the desired data, download it and process it. The data was gathered in the laptop

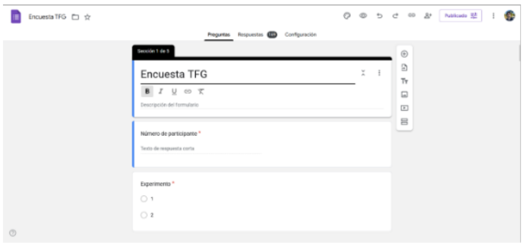

Figure 6.3: Google Forms Example

of my proffesor as they have the paid license.Some of the most interesting tools are the visualization of heat maps, recordings and the Areas of Interest, that will be further explained.

6.3.3 Google Forms

Google Forms is a free tool developed by Google that allows google users to create forms. This is a powerful tool as the answers can be stored in a Google Spreadsheet that can be exported as a .csv archive to be later processed. This tool is also pretty compatible with the tobii pro lab when creating the timelines that the user is going to be exposed due to the fact of the possibility of running.

6.3.4 Python 3

There're few programming languages that almost everyone on earth has heard of, this is the case. Python is a powerful programming language not because of its speed or performance but because of the libraries that are available. This python environment makes room to one of the biggest spaces for data processing. In my case, I'm not a specialized student on data processing, that's why a well-known and big space like this can be the perfect tool to approach a thesis.

6.3.5 Dark Reader

Given that the objective is to analyze the effects of display polarity on texts the decision was to choose a wiki page as it is a tool that until this day many users do. Moreover, an e-commerce is another safe bet. In order to make it the most average the decision was to use "wikiedia.org" and "amazon.es". However, the e-commerce website didn't allow negative polarity, so we had to use a firefox extension called Dark Reader which you can see in figure 6.4

Figure 6.4: Dark Reader Extension[1]

Chapter 7

Experiment Execution

7.1 Environmental Conditions

The experiment was developed in the Faculty of Economics of the University of Oviedo, located in the city of Oviedo during the 23rd and 25th of April. The weather was cloudy with few intermittent rains, this led to a general dim light. In order to have proper lighting and stable conditions the room was illuminated with fluorescent lights giving a total of 7,2 Exposure Value (EV) on the first day and 7,3 EV the second day.

Regarding the placement of the experiment, we were in a Main Hall in the calmest possible place. Experiment subjects would sit in a chair with a small table holding the laptop and the eye-tracker. The average distance between the experiment subjects and the screen was 60 cm as suggested by the tobii pro lab.

7.2 Experiment Phases

The first and most 3 phase of the experiment consisted of a tobii calibration. As this is a key factor and it was desired to have a lot of precision during the experiment, that is why the 9-point calibration was chosen. Experiment subjects were asked to watch and follow a white dot on the screen after this was finished the calibration results were shown. In the case the results were favorable the experiment would continue or else the calibration would be repeated entirely or partially.

7.2.1 Eye-tracker Calibration

The second phase of the experiment already had two different versions. One consisted of the instructions displayed in negative polarity and the other one prompted the instructions in positive polarity. Results of this have been recorded but not taken into account inside the experiment as they are outside of the aim of web based texts.

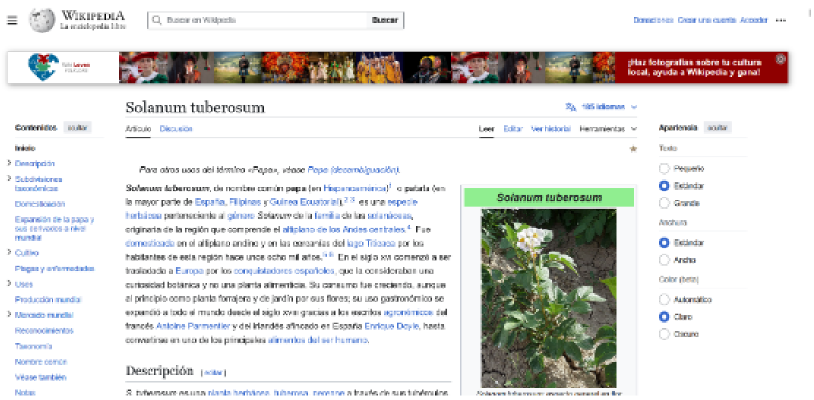

Figure 7.1: Wikipedia page about potatoes in positive display polarity[14]

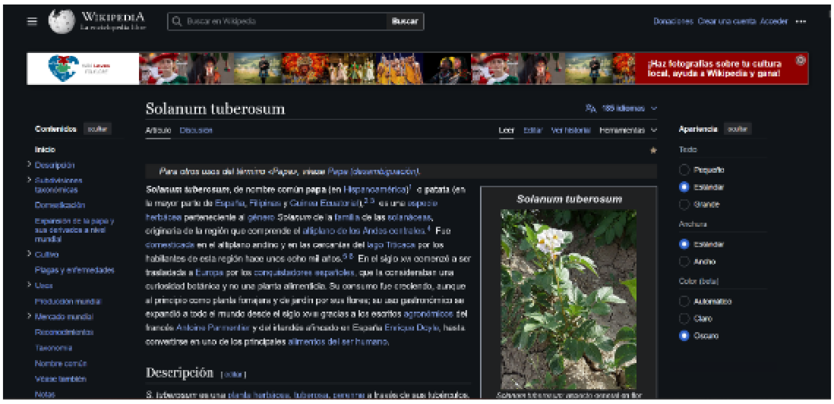

Figure 7.2: Wikipedia page about potatoes in negative display polarity[14]

7.2.2 User Instructions

7.2.3 Reading Task

The reading task consisted of 2 texts. The first texts is based on Wikipedia newest negative and positive display polarity while the first text is based on Amazon web-page along with the plugin "Dark Reader" explained above in the document. The pages displayed were centered around daily objects with complex enough information so It wouldn't be a technical hurdle while not being too simple to read.

Wikipedia article

The chosen Wikipedia page refers to such a common food as the potato A CONTINUAR AQUI!!!!!!!!!!!!!!!!!!!!

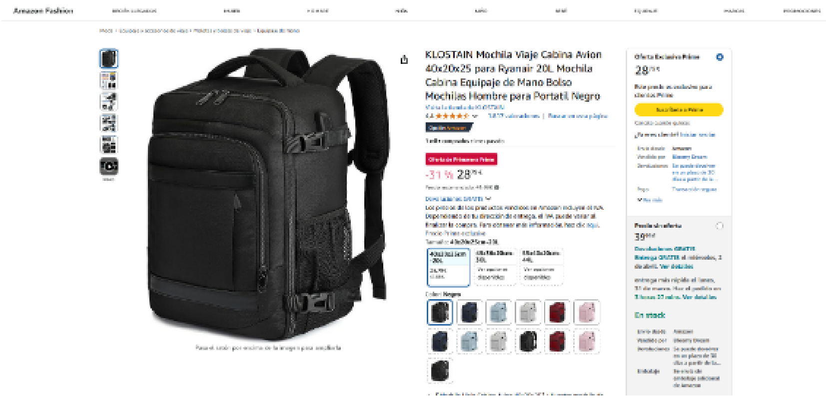

Figure 7.3: Amazon product page in positive display polarity[6]

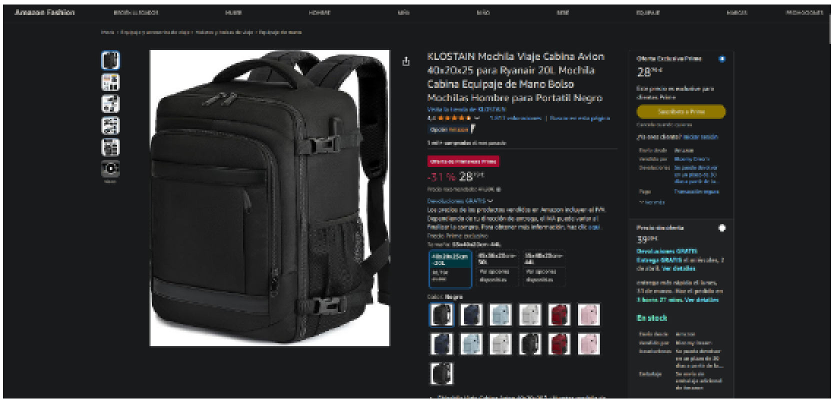

Figure 7.4: Amazon product page in negative display polarity[6]

Amazon product

7.2.4 Final Questionnaire

Chapter 8

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8.1 Data Preprocessing

8.2 Chosen Metrics

8.3 Statistical Analysis Methods

Chapter 9

Results

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9.2 Questionnaire Results

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Chapter 10

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Chapter 11

Conclusion and Future Work

11.1 Lessons Learned Report

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Appendix A

Project Management

A.1 Initial Planning

A.2 Risk Identification and Management

Risk identification is a key phase of the project as it can help acknowledge possible negative effects on the project. It is important to have a strategy to face such events in order to make them affect the least possible the outcome of the project.

Table A.1: Risks List

Risk ID	Risk Name
1	Few Experimental Samples
2	Statistical Analysis Imprecision
3	Loss of Partial Work
4	Justified Leave of Worker
5	Lack of Teamwork
6	Tobii Measurements Drifts
7	Privacy Violations
8	Timeline Delay
9	Publisher Rejection

A.2.1 Risk Details

Few Experimental Samples

There is not enough sample for the test, this can lead to inconsistent data and problems when doing the processing and publishing the paper.

Statistical Analysis Imprecision

When analyzing the processed data if the statistic analysis is imprecise it can lead to wrong conclusions. It is important for it to remain precise.

Table A.2: Few experimental samples impact.

Probability	Impact				Impact
	Budget	Planning	Reach	Quality	
Medium	Low	Low	Medium	Critic	0,45

Table A.3: Statistical analysis imprecision impact.

Probability	Impact				Impact
	Budget	Planning	Reach	Quality	
Low	Seamless	Low	Low	Medium	0,09

Loss of Partial Work

Due to data corruption or any other external agent, there's a possibility of losing the entire or part of the work accomplished.

Table A.4: Loss of partial work impact.

Probability	Impact				Impact
	Budget	Planning	Reach	Quality	
Low	Medium	High	Medium	Low	0,17

Justified Leave of Worker

Any of the workers may need a justified leave of work according to the Spanish law.

Lack of Teamwork

Teamwork is important as it creates a healthy environment where workers feel comfortable, however it is quite easy to have a lot of individualism in teams.

Table A.5: Justified leave of worker impact.

Probability	Impact				Impact
	Budget	Planning	Reach	Quality	
Medium	Low	High	Low	Low	0,28

Table A.6: Lack of teamwork impact.

Probability	Impact				Impact
	Budget	Planning	Reach	Quality	
Medium	Low	High	Low	Low	0,15

Tobii Measurements Drifts

Due to light changes, the subject of the experiment looking away or other uncontrollable events, the tobii eye-tracker may drift its measurements.

Table A.7: Tobii measurements drifts impact.

Probability	Impact				Impact
	Budget	Planning	Reach	Quality	
High	Seamless	Seamless	Seamless	Medium	0,21

Privacy Violations

This whole research gathers a lot of sensitive information and it is a key to store and manipulate the sensitive data correctly as it can lead to fines.

Timeline Delay

Research projects usually take longer than initially expected this can overextend the planning making it a real risk.

Table A.8: Privacy violations impact.

Probability	Impact				Impact
	Budget	Planning	Reach	Quality	
Low	Critic	Low	Seamless	Low	0,27

Table A.9: Timeline delay impact.

Probability	Impact				Impact
	Budget	Planning	Reach	Quality	
High	High	High	Medium	Low	0,39

Publisher Rejection

After finishing the experiment, gathering the data and writing the paper there's a possibility the desired scientific publishers may not be interested in the paper.

Table A.10: Publisher rejection impact.

Probability	Impact				Impact
	Budget	Planning	Reach	Quality	
High	Seamless	Low	High	Low	0,39

A.2.2 Risk Strategy

A risk strategy is needed in order to know what to do against every risk, the risks of A.2.1 explains them further. From the 9 identified risks, 2 will be deleted, 3 assumed and 4 mitigated.

Table A.11: Risk strategy and action plan.

ID	Risk Name	Impact	Strategy	Risk Answer
1	Few Experimental Samples	0,45	Delete	Spend more time than planned gathering samples for the experiment.
8	Timeline Delay	0,39	Mitigate	When creating the planning always adding more time than expected.
9	Publisher Rejection	0,39	Assume	It should be necessary to have a broad list of publishers.
4	Justified Leave of Worker	0,28	Assume	Work the planning around the person missing.
7	Privacy Violations	0,27	Delete	Exhaustive analysis by the prepared workers on the RGPD.
6	Tobii Measurements Drifts	0,21	Assume	Try to find a spot with balanced lighting.
3	Loss of Partial Work	0,17	Mitigate	Integrate a version control system on the work.
5	Lack of Teamwork	0,15	Mitigate	Weekly optional team buildings outside of the work period.
2	Statistical Analysis Imprecision	0,09	Mitigate	Final deep review of the statistical analysis.

A.3 Budget and Resource Allocation

A.4 Project Monitoring and Incidence Log